# EXAMPLE 8

## Two-Story, Three-Dimensional Frame - Dynamic Response Spectrum Analysis

## Problem Description

This is a two-story, three-dimensional building frame subjected to a response spectrum of constant amplitude. The three-dimensional structure is modeled as a single frame with nine column lines and twelve bays. Kip-foot-second units are used.

For consistency with the models documented in other computer programs with which the ETABS results are compared (see Table 8-1), no story mass moments of inertia are assigned in the ETABS model.

## Geometry, Properties and Loads

The geometry of the structure is shown in Figure 8-1.

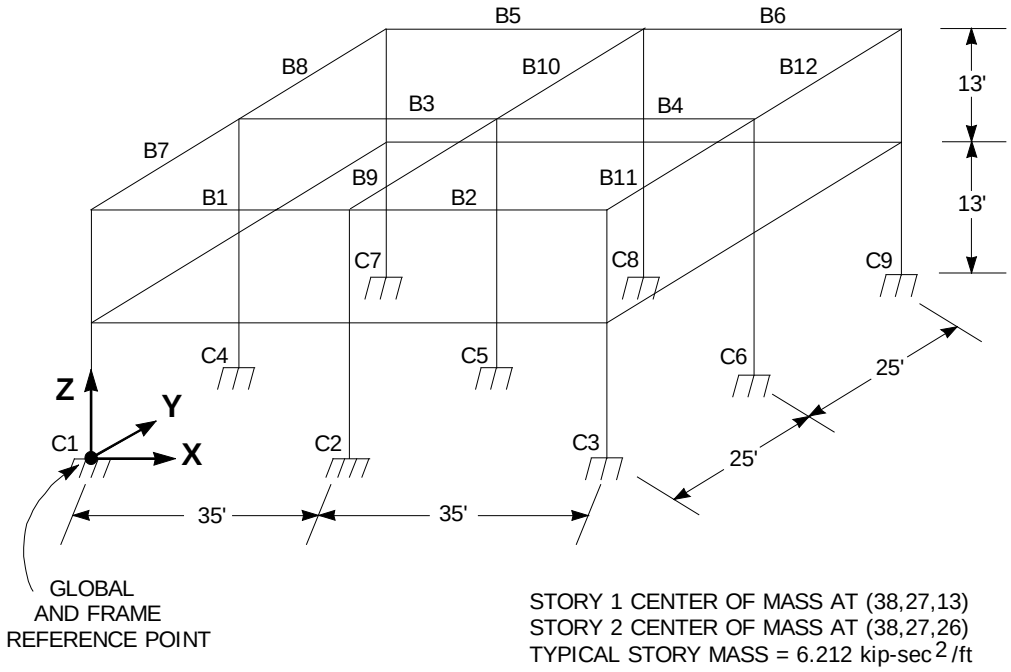


***Figure 8-1 Two-Story Three-Dimensional Frame***

A response spectrum with a constant value of 0.4g is used. Other parameters associated with the structure are as follows:

| | Columns | Beams |
|---|---|---|
| Axial area | 4 $ft^2$ | 5 $ft^2$ |
| Minor moment of inertia | 1.25 $ft^4$ | 1.67 $ft^4$ |
| Major moment of inertia | 1.25 $ft^4$ | 2.61 $ft^4$ |
| Modulus of elasticity | 350000 ksf | 500000 ksf |

## Technical Features of ETABS Tested

- Three-dimensional frame analysis
- User-specified section properties
- Dynamic response spectrum analysis

## Comparison of Results

This example is also analyzed in Wilson and Habibullah (1992) and Peterson (1981). A comparison of the key ETABS results with Wilson and Habibullah (Reference 1) and Peterson (Reference 2) is shown in Table 8-1.

**Table 8-1 Comparison of Results**

| Quantity | ETABS | Reference 1 | Reference 2 |
|---|---|---|---|
| Period, Mode 1 | 0.22708 | 0.22706 | 0.22706 |
| Period, Mode 2 | 0.21565 | 0.21563 | 0.21563 |
| Period, Mode 3 | 0.07335 | 0.07335 | 0.07335 |
| Period, Mode 4 | 0.07201 | 0.07201 | 0.07201 |
| X-Displacement Center of mass, 2nd Story | 0.0201 | 0.0201 | 0.0201 |

## Computer File

The input data file is Example 08.EDB. This file is provided as part of the ETABS installation.

## Conclusion

The results comparison shows acceptable agreement between ETABS and the references.